

Durability Assessment of Glass FRP Solid and Hollow Bars (Rock Bolts) for Application in Ground Control of Jurong Rock Caverns in Singapore

Brahim Benmokrane¹; Mathieu Robert²; Hamdy M. Mohamed³; Ahmed H. Ali⁴; and Patrice Cousin⁵

Abstract: This study was conducted to investigate the durability of two types of vinyl-ester/glass fiber-reinforced polymer (GFRP) rock bolts [solid and hollow (tubular) GFRP bars] that were subsequently used for ground control of the Jurong rock caverns in Singapore. The testing program aimed to assess the long-term durability characteristics of unstressed hollow and solid GFRP rock bolts subjected to a saline solution (simulating the subsea cavern water of the Jurong rock caverns in Singapore) and predicting their long-term tensile strength in such environments. The study simulated rather aggressive conditions by immersing the hollow and solid GFRP bars (rock bolts) in the solution at different elevated temperatures (20, 40, and 50°C) in climate chambers for different exposure times of 1,000, 3,000, and 5,000 h. Arrhenius plots were applied to the test results, extrapolating the project's life to 50 years at a mean annual temperature (MAT) of 32°C. The predicted tensile strength at a MAT of 32°C is 96.8 and 97.3% for a service life of 50 years for solid and hollow GFRP bars, respectively. Based on the obtained results, the tested GFRP rock bolts were recommended as alternatives to stainless-steel rock bolts and successfully used as ground control in the Jurong rock caverns in Singapore. DOI: 10.1061/(ASCE)CC.1943-5614.0000775. © 2016 American Society of Civil Engineers.

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Introduction

Recently, fiber-reinforced polymer (FRP) bars have come on the market for use as rock bolts as alternatives to conventional ground-control elements made of steel (Zhang et al. 2001). Indeed, several studies have addressed the durability of glass FRP (GFRP) bars as internal reinforcement of concrete (ACI 2015). Very limited studies, however, have been conducted on the durability of GFRP bars used as rock bolts and subjected to saline solutions or to simulated mining environments, which would be of crucial relevance to the Jurong rock caverns project. The properties of the pultruded products (void content, interface quality, and fiber distribution) and the nature of the aggressive environment can lead to important changes in the final product's behavior and durability. Therefore, the available data on the durability of GFRP reinforcing bars

subjected to typical alkaline may not be directly applicable to GFRP rock bolts subjected to saline solution and high temperature.

Jurong rock caverns (JRC), Southeast Asia's first underground rock-cavern facility for storing liquid hydrocarbons, is located at a depth of 150 m below Jurong Island's Banyan Basin in Singapore. Early in the design process, conventional steel rock bolts with resin polymer grout were considered for ground control of the caverns. Owing to the corrosive hydrologic environment, weight and handling considerations, local construction preferences, and the requirement to have a cavern design life of 50 years, GFRP rock bolts were proposed as an alternative to stainless-steel bolts. More than 80,000 cement-grouted GFRP solid and tubular bars (similar to those tested in this study) varying in length from 3.5 to 5.5 m were used in this project.

To assess the feasibility of using GFRP bolts in the Jurong rock caverns in Singapore, an elaborate test program was carried out at the University of Sherbrooke in collaboration with Hyundai Engineering and Construction (Singapore) and Firep International AG (Switzerland) (Benmokrane et al. 2012). The test program included two types of vinyl-ester/glass fiber-reinforced polymer (GFRP) rock bolts consisting of solid and hollow GFRP bars (with hollow-core). The purpose was to assess the environmental durability of the solid and hollow GFRP bars subjected to a saline solution, simulating the subsea cavern water and predicting their long-term behavior in such environments to determine if they were appropriate for use in the Jurong rock caverns. Tensile strength was considered the primary structural parameter and was therefore used an indicator of degradation attributable to exposure. In addition, the Arrhenius theory was used to predict the durability and service life of the new GFRP bars used as rock bolts for ground control such as in mining environments or as reinforcement for concrete structures.

¹Professor of Civil Engineering and Tier-1 Canada Research Chair in Advanced Composite Materials for Civil Structures and NSERC Chair in FRP Reinforcement for Concrete Structures, Dept. of Civil Engineering, Univ. of Sherbrooke, QC, Canada J1K 2R1 (corresponding author). E-mail: Brahim.Benmokrane@usherbrooke.ca

²Assistant Professor, Dept. of Civil Engineering, Univ. of Sherbrooke, QC, Canada J1K 2R1. E-mail: Mathieu.Robert2@USherbrooke.ca

³Research Associate, Dept. of Civil Engineering, Univ. of Sherbrooke, QC, Canada J1K 2R1; Assistant Professor, Helwan Univ., Egypt. E-mail: Hamdy.Mohamed@usherbrooke.ca

⁴Ph.D. Candidate, Dept. of Civil Engineering, Univ. of Sherbrooke, QC, Canada J1K 2R1. E-mail: Ahmed.Ali@usherbrooke.ca

⁵Research Associate, Dept. of Civil Engineering, Univ. of Sherbrooke, QC, Canada J1K 2R1. E-mail: Patrice.Cousin@USherbrooke.ca

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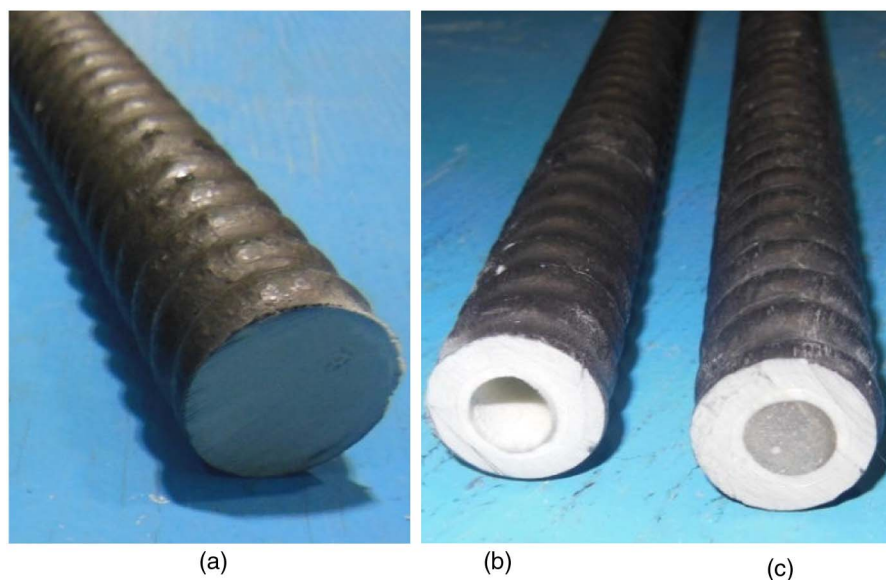


Fig. 1. GFRP rock-bolt specimens: (a) solid bar; (b) hollow bar; (c) hollow bar filled with cement grout

Experimental Approach

Solid and Hollow GFRP Rock Bolts

Solid (Type K60/25) and hollow [tubular-Type J64/28/(12)] GFRP rock bolts manufactured by Firep International AG (Switzerland) were examined and tested at the University of Sherbrooke and, as a result, were subsequently used as rock bolts (passive anchors) for ground control of the JRC in Singapore. A passive soil anchor is tensioned as the structure itself applies load to it. The designed load for these GFRP anchors is less than 25% of the guaranteed tensile strength (code stress limit). Past research indicated that this stress level (25%) has no significant effect on the durability and long-term prediction of GFRP bars (Robert and Benmokrane 2010). The solid GFRP bolts (K60/25) had an outer diameter of 25 mm, whereas the hollow bolt [J64/28/(12)] had an outer diameter of 28 mm with an inner diameter of 12 mm. Rock bolts are named in the tunneling industry by their outer diameter, whereas rebars for concrete structures are referred to by their nominal diameter. The solid and hollow GFRP bolts were designed to have the same cross-sectional area for stress calculation: 350 mm² as the nominal area and 443 mm² as the effective cross-sectional area (area by immersion test). All the tensile properties reported in this paper were determined using the nominal area of 350 mm² for both GFRP rock bolts. Fig. 1 shows the solid and hollow GFRP bolts, which have continuously profiled surfaces. Both the solid and hollow GFRP bolts were made of continuous ECR-glass (special-purpose glass fibers with high corrosion resistance) fibers impregnated in a vinyl-ester resin using the pultrusion process, leading to a cure ratio of more than 99%. The mass fractions of glass were 83.0 and 82.5% for the solid and hollow bolts, respectively, and were determined by thermogravimetric analysis according to ASTM (2003). Their relative densities according to ASTM (2012) were 2.11 and 2.09 for the solid and hollow bolts, respectively. All the bars were cut into 1,440-mm lengths to perform tensile testing as specified in ASTM (2011). The GFRP bars or bolts were divided into three series: (1) the unconditioned reference samples (solid and hollow); (2) the conditioned solid rock-bolt samples (27 bars) immersed in saline solution; and (3) conditioned hollow rock-bolt samples (27 bars) filled with cement grout and immersed in saline solution.

The cement grout used to fill the hollow GFRP bars consisted of Type I cement according to ASTM (2005) and had a water-to-cement ratio (w/c) of 0.40.

Test Plan

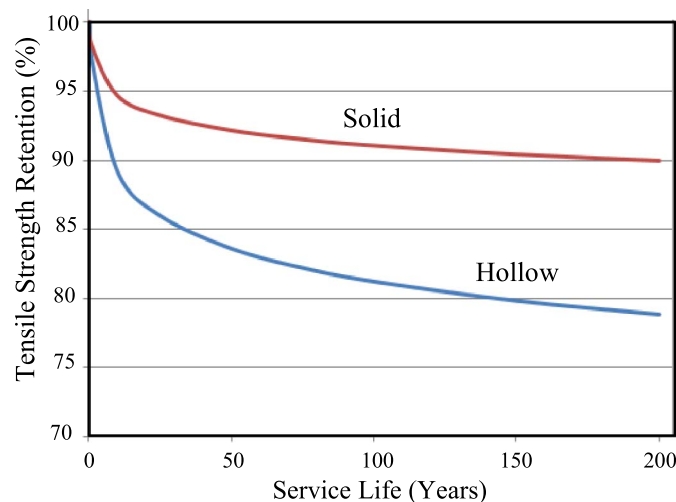
This study used accelerated aging in a saline solution simulating the subsea cavern water in the JRC project, where the rock bolts were subsequently used as a result of this research. The specimens were completely immersed at three different temperatures (23, 40, and 50°C) and were removed from the water after three different periods of time (1,000, 3,000, and 5,000 h) to lead to predictions about long-term properties. The alkalinity of the saline solution was based on the collected rock water from the Jurong rock caverns (Singapore), which was evaporated by 20% in volume to increase the concentration. The water characteristic (in terms of salt and pH-value) of the job site was reproduced at the laboratory. An average value of the concentration of anions and cations in the mine water was determined in collaboration with the Department of Chemistry at the University of Sherbrooke. The pH value of the solution was around 6.7. Table 1 presents the complete composition of the saline solution used in this study. The pH of the solution surrounding the cement grout-filled hollow bolts resulted from the pH of the saline solution and from the cement grout absorbing water, thereby releasing alkaline ions into the bar environment. All the GFRP rock bolts were tested under tension according to ASTM (2011). The conditioned test specimens were washed in water, dried with tissue paper, and tensile tests were carried out directly

Table 1. Complete Composition of the Saline Solution Used in This Study

Compound	Concentration (g/L of distilled water)	Ionic content	
		Ion	Content (ppm)
Sodium chloride (NaCl)	26	Cl ⁻	16,760
Potassium sulfate (K ₂ SO ₄)	0.6	SO ₄ ²⁻	2,330
Calcium chloride (CaCl ₂)	1.5	Na ⁺	10,200
Magnesium sulfate (Mg ₂ SO ₄)	3.0	Mg ²⁺	1,000
—	—	Ca ²⁺	540
—	—	K ⁺	270

Table 2. Test Matrix and Longitudinal Tensile Properties of Reference and Conditioned Specimens

Immersion time (h)	Temperature (°C)	Number of specimens	Mean tensile strength (MPa)				Modulus of elasticity (GPa)				Ultimate strain (%)			
			Solid		Hollow		Solid		Hollow		Solid		Hollow	
			COV (%)		COV (%)		COV (%)		COV (%)		COV (%)		COV (%)	
0	23	5	1,069 ± 33	3	1,042 ± 82	8	60.8 ± 1.7	2.8	62.8 ± 1.5	2.3	1.76 ± 0.1	4.6	1.66 ± 0.1	5.9
1,000	23	3	1,066 ± 29	3	1,084 ± 21	2	55.1 ± 3.3	5.9	63.4 ± 0.7	1.1	1.94 ± 0.1	4.1	1.71 ± 0.1	1.5
	40	3	1,037 ± 41	4	1,045 ± 7	1	56.1 ± 0.9	1.5	63.4 ± 1.1	1.7	1.85 ± 0.1	4.3	1.65 ± 0.1	1.8
	50	3	1,047 ± 21	2	1,006 ± 152	5	59.6 ± 0.9	1.6	62.0 ± 1.0	1.4	1.76 ± 0.1	3.54	1.62 ± 0.2	1.4
3,000	23	3	1,053 ± 40	4	1,059 ± 5	1	57.4 ± 2.9	5.0	62.2 ± 0.6	0.9	1.84 ± 0.1	4.9	1.70 ± 0.1	1.0
	40	3	1,023 ± 29	3	1,051 ± 93	9	56.1 ± 1.7	3.0	61.9 ± 1.4	2.2	1.82 ± 0.1	4.2	1.70 ± 0.1	6.6
	50	3	1,029 ± 30	4	977 ± 43	4	59.3 ± 0.9	1.5	60.5 ± 1.9	3.1	1.74 ± 0.1	4.4	1.61 ± 0.1	1.3
5,000	23	3	1,084 ± 18	2	1,023 ± 72	7	57.5 ± 2.9	5.0	61.9 ± 1.8	3.0	1.89 ± 0.1	3.33	1.65 ± 0.1	4.6
	40	3	1,012 ± 28	3	965 ± 16	2	57.8 ± 2.6	4.4	62.1 ± 0.7	1.1	1.76 ± 0.1	7.10	1.56 ± 0.1	2.6
	50	3	1,017 ± 3	1	927 ± 77	8	58.7 ± 1.8	3.0	62.9 ± 1.1	1.7	1.73 ± 0.1	2.73	1.47 ± 0.1	6.8

**Fig. 2.** General relationship between the PR and the predicted service life at a mean annual temperature of 32°C (Singapore)

so soon after conditioning that drying was precluded. The hollow GFRP rock bolts were tested with their cement-grout core centered to avoid any damage to the specimen resulting from the removal of the cement grout.

Tests Results and Discussion

Tensile-Strength Retention

The tensile testing of the unconditioned specimens showed an approximately linear behavior up to failure. All the specimens failed suddenly, as expected, as the result of fiber rupture. A similar mode of failure was observed for the specimens immersed in the saline solution. Notably, no chemical deposits were observed on the surface of the conditioned GFRP bars before tensile testing for residual properties.

Table 2 shows the experimental results obtained during the tensile tests concerning the longitudinal tensile properties of the solid and hollow aged bars tested after immersion. The slight variation in tensile strength is probably related to the slight increase in moisture absorption over time, leading to plasticizing effects of the polymer matrix. Moreover, the test results in this study indicate that the degradation rate stabilized and decreased quickly as the exposure time increased from 3,000 to 5,000 h.

Prediction of Long-Term Behavior

The preexposure and postexposure tensile strengths of the GFRP bars were used for long-term property predictions based on the Arrhenius theory (Benmokrane et al. 2016; Bank et al. 2003). Consequently, calculation was made for the predicted tensile-strength retention (PR) as a function of time for immersion at 32°C, and the general relationship between the PR and the predicted service life at the average temperature of 32°C can be drawn (Fig. 2). The predicted time to reach the determined tensile-strength retention level (PR) for the solid and tubular GFRP rock bolts aged is approximately 150 years for a PR of more than 90 and 80%, respectively. These predictions show that the solid and tubular GFRP rock bolts tested in this study are durable with respect to the saline environment encountered in mining applications, because the saline solution used in this study closely simulates the subsea cavern water. As a result, they could be adopted for the Jurong rock caverns project.

Conclusions

In this research study, unstressed solid and hollow GFRP bars (rock bolts) were exposed to a saline solution simulating subsea cavern water at 23, 40, and 50°C to accelerate the effect of the mining environment. The preexposure and postexposure tensile strengths of the GFRP bars were deemed indicative of specimen durability and were used for long-term property predictions based on the Arrhenius theory. According to the experimental results and long-term predictions, even after a service life of 100 years, the tensile-strength retention of the tested GFRP rock bolts would still be greater than 82% for a mean annual temperature of 32°C (representative of Singapore).

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